

DIVISORS IN WINDOWS AND THE EULER–MASCHERONI CONSTANT

DAVID V. FELDMAN

ABSTRACT. For $0 < c < 1$ the number of divisors of q below q^c averages, over $q \leq B$, to $c \log B + (\gamma - c) + O(B^{-\min(c, 1-c)})$, where γ denotes the Euler–Mascheroni constant. Consequently the number of divisors in a window (q^a, q^b) averages to $(b-a)(\log B - 1) + o(1)$: the constant γ cancels, and the average counts for two windows stand asymptotically in the ratio of the windows’ exponent lengths. Anchoring a window at a threshold and its square root makes the logarithms cancel instead: for weakly increasing F with $F(q) \rightarrow \infty$ and $q/F(q) \rightarrow \infty$, a number q has, on average, exactly γ more divisors below $\sqrt{F(q)}$ than in $[\sqrt{F(q)}, F(q))$. For windows on the scale of q itself the average depends on an endpoint convention: with $\alpha \in (0, 1)$, counting divisors below $\sqrt{\alpha q}$ against divisors in $(\sqrt{\alpha q}, \alpha q)$ yields the average $H_{\lfloor 1/\alpha \rfloor} - \log(1/\alpha)$, while the window $(\sqrt{\alpha q}, \alpha q]$ yields $H_{\lfloor 1/\alpha \rfloor - 1} - \log(1/\alpha)$; the two constants differ exactly when $1/\alpha$ takes an integer value, and then by α . Either function of α oscillates about γ , tends to γ as $\alpha \rightarrow 0$, and integrates over $(0, 1)$ to $\zeta(2) - 1$.

1. INTRODUCTION

The Euler–Mascheroni constant

$$\gamma = \lim_{n \rightarrow \infty} \left(\sum_{k=1}^n \frac{1}{k} - \log n \right) = 0.5772156649 \dots$$

admits few interpretations that avoid explicit mention of logarithms. Among the candidates collected by Finch [2], the most arithmetical goes back to de la Vallée Poussin [1]: if a large integer n is divided by each $k \leq n$, the fractions $\{n/k\}$ by which the quotients fall short of the next integer average not to $1/2$ but to $1 - \gamma$.

This paper locates γ inside a family of statements about counting divisors in intervals, and separates two regimes. In the first regime the interval endpoints grow like fixed powers $q^a < q^b$ of the number q whose divisors we count. Here the average count obeys a law from which γ is absent:

the average number of divisors of $q \leq B$ in the window (q^a, q^b) equals $(b - a)(\log B - 1) + o(1)$.

In particular the average counts attached to two windows stand asymptotically in the ratio of the windows’ exponent lengths $b - a$; for example, a number has on average twice as many divisors in $(n^{1/4}, n^{1/2})$ as in $(n^{1/8}, n^{1/4})$ (Corollary 6).

In the second regime an interval anchors at a threshold $F(q)$ and its square root, with $F(q)$ growing, but growing more slowly than q . Subtracting the count above $\sqrt{F(q)}$ from the count below makes the $\log B$ terms cancel and leaves the constant (Theorem 9):

on average, q has exactly γ more divisors below $\sqrt{F(q)}$ than between $\sqrt{F(q)}$ and $F(q)$.

2020 *Mathematics Subject Classification.* 11N37; 11Y60.

Key words and phrases. Euler–Mascheroni constant, divisors, average order, harmonic numbers.

Taking $F(q) = \sqrt{q}$ gives the most quotable instance: γ measures the average excess of divisors below the fourth root over divisors between the fourth root and the square root.

The hypothesis $q/F(q) \rightarrow \infty$ cannot be dropped. At the extreme $F(q) = q$ the divisors of q pair off as $d \leftrightarrow q/d$ across $\sqrt{q}$, so every q has *exactly* as many divisors below its square root as above, and the discrepancy vanishes identically rather than averaging to γ . Between the two regimes sit windows on the scale of q itself, from $\sqrt{\alpha q}$ to αq for fixed $\alpha \in (0, 1)$. The limiting discrepancy then equals $H_K - \log(1/\alpha)$, where H_K denotes a harmonic number whose index depends on α and on the window's endpoint convention: when $1/\alpha$ takes an integer value k , the point $d = \alpha q$ is itself a divisor of q for one q in every k , so including or excluding the right endpoint shifts the average by α . Theorem 13 records both constants. As α decreases to 0 the average oscillates about γ and converges to γ , and its mean over $\alpha \in (0, 1)$ equals $\zeta(2) - 1$ (Proposition 15).

All proofs are elementary, resting on the divisor–codivisor correspondence $q = de$ and on quantitative forms of the definition of γ ; nothing below requires more than the Dirichlet divisor asymptotic, itself derived in passing. Section 2 fixes notation and records the counting and harmonic estimates. Section 3 proves the threshold and window laws. Section 4 proves the γ -discrepancy theorem for general F . Section 5 treats windows on the scale of q and the mean value $\zeta(2) - 1$. Section 6 isolates a logarithm-free approximation to γ implicit in the earlier arguments and computes its rate: the error after the n th stage equals $\frac{2}{3}n^{-2} + O(n^{-4})$. Section 8 states open questions.

2. PRELIMINARIES

Throughout, q, d, e, k, m, n denote positive integers and B a positive integer tending to infinity. For an interval $I \subseteq \mathbb{R}$ put

$$\tau(q; I) = \#\{d : d \mid q, d \in I\}, \quad \tau(q) = \tau(q; [1, q]),$$

and write $H_m = \sum_{k=1}^m 1/k$ (with $H_0 = 0$) for the harmonic numbers. Implied constants may depend on fixed parameters such as c, a, b, α , but never on B .

The proofs repeatedly exchange a divisor d of q for its codivisor $e = q/d$.

Lemma 1 (counting by codivisors). *For any predicate P on pairs of positive integers,*

$$\sum_{q \leq B} \#\{d \mid q : P(d, q/d)\} = \#\{(d, e) : de \leq B, P(d, e)\}.$$

In particular, for any $S \subseteq \mathbb{N}$,

$$\sum_{q \leq B} \#\{d \mid q : d \in S\} = \sum_{\substack{d \in S \\ d \leq B}} \left\lfloor \frac{B}{d} \right\rfloor.$$

Proof. The pairs (q, d) with $d \mid q \leq B$ correspond to the pairs (d, e) with $q = de$. The second display counts, for each $d \in S$, the multiples of d up to B . $\square$

Lemma 2 (harmonic estimates). *For $m \geq 1$,*

$$H_m = \log m + \gamma + \theta_m, \quad 0 < \theta_m < \frac{1}{m}.$$

Moreover $H_m = \log m + \gamma + \frac{1}{2m} - \frac{1}{12m^2} + O(m^{-4})$.

Proof. The sequence $u_m = H_m - \log m$ strictly decreases, since $u_m - u_{m+1} = \log(1 + 1/m) - 1/(m+1) > 0$, and converges to γ by definition of γ ; hence $\theta_m = u_m - \gamma > 0$. For $n > m$,

$$u_m - u_n = \sum_{k=m+1}^n \left(\log \frac{k}{k-1} - \frac{1}{k} \right) < \sum_{k=m+1}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) < \frac{1}{m},$$

and letting $n \rightarrow \infty$ gives $\theta_m \leq 1/m$, with equality excluded by strictness. The refined expansion is Euler–Maclaurin summation applied to H_m ; see [5, §9.6]. $\square$

Lemma 3 (Dirichlet). $\sum_{q \leq B} \tau(q) = B \log B + (2\gamma - 1)B + O(\sqrt{B})$.

Proof. By Lemma 1 the sum counts the lattice points under the hyperbola $de \leq B$. Counting the points with $d \leq \sqrt{B}$, the points with $e \leq \sqrt{B}$, and subtracting the doubly counted square gives

$$\sum_{q \leq B} \tau(q) = 2 \sum_{d \leq \sqrt{B}} \left\lfloor \frac{B}{d} \right\rfloor - \lfloor \sqrt{B} \rfloor^2 = 2B H_{\lfloor \sqrt{B} \rfloor} - B + O(\sqrt{B}),$$

and Lemma 2 completes the computation. $\square$

3. THRESHOLDS AT FIXED POWERS, AND THE WINDOW LAW

Theorem 4 (threshold law). *Fix $c \in (0, 1)$. Then*

$$\sum_{q \leq B} \tau(q; [1, q^c]) = cB \log B + (\gamma - c)B + O(B^{\max(c, 1-c)}),$$

so the number of divisors below q^c averages to $c \log B + (\gamma - c) + O(B^{-\min(c, 1-c)})$.

Proof. Write $\kappa = (1 - c)/c$, so that $\kappa + 1 = 1/c$. For a pair (d, e) with $q = de$,

$$d < q^c \iff d^{1/c} < de \iff e > d^\kappa.$$

By Lemma 1 the sum in question counts the pairs (d, e) with $d^\kappa < e \leq B/d$. If $d > B^c$ then $d^\kappa = d^{1/c}/d > B/d$ and no e qualifies. If $d \leq B^c$ then $d^\kappa \leq B/d$, and the number of qualifying e equals $\lfloor B/d \rfloor - \lfloor d^\kappa \rfloor = B/d - d^\kappa + O(1)$. With $D = \lfloor B^c \rfloor$ the sum therefore equals

$$\sum_{d \leq D} \left(\frac{B}{d} - d^\kappa \right) + O(D) = B H_D - \sum_{d \leq D} d^\kappa + O(B^c).$$

Since $D = B^c + O(1)$, Lemma 2 gives

$$B H_D = B \log D + \gamma B + O(B^{1-c}) = cB \log B + \gamma B + O(B^{1-c}),$$

using $\log D = c \log B + O(B^{-c})$. Comparison with $\int_0^D x^\kappa dx$ gives

$$\sum_{d \leq D} d^\kappa = \frac{D^{\kappa+1}}{\kappa+1} + O(D^\kappa) = c D^{1/c} + O(B^{1-c}) = cB + O(B^{1-c}),$$

since $D^{1/c} = (B^c + O(1))^{1/c} = B + O(B^{1-c})$ and $D^\kappa \leq B^{1-c}$. Assembling the pieces yields the statement. $\square$

Corollary 5 (window law). *Fix $0 < a < b < 1$. Then*

$$\sum_{q \leq B} \tau(q; [q^a, q^b]) = (b - a) B (\log B - 1) + O(B^{\max(b, 1-a)}),$$

and the same asymptotic holds for each of the four endpoint conventions.

Proof. Since $q^a \leq q^b$, $\tau(q; [q^a, q^b]) = \tau(q; [1, q^b]) - \tau(q; [1, q^a])$; apply Theorem 4 at b and at a and subtract. The γ terms cancel. As for endpoints: a divisor $d = q^a$ corresponds under $q = de$ to a pair with $e = d^{(1-a)/a}$ exactly, hence to at most one pair for each $d \leq B^a$, so the boundary contributes $O(B^a)$, within the stated error; likewise at q^b . $\square$

Corollary 6 (ratio law). *Fix $0 < a < b < 1$ and $0 < u < v < 1$. Then*

$$\frac{\sum_{q \leq B} \tau(q; (q^a, q^b))}{\sum_{q \leq B} \tau(q; (q^u, q^v))} \longrightarrow \frac{b - a}{v - u} \quad (B \rightarrow \infty).$$

In particular, on average a number has twice as many divisors in $(n^{1/4}, n^{1/2})$ as in $(n^{1/8}, n^{1/4})$; indeed

$$\frac{1}{B} \sum_{q \leq B} \left(\tau(q; (q^{1/4}, q^{1/2})) - 2 \tau(q; (q^{1/8}, q^{1/4})) \right) = O(B^{-1/8}).$$

Proof. Immediate from Corollary 5. In the displayed instance the main terms $\frac{1}{4}(\log B - 1)$ and $2 \cdot \frac{1}{8}(\log B - 1)$ cancel, and the error terms are $O(B^{\max(1/2, 3/4)-1}) + O(B^{\max(1/4, 7/8)-1}) = O(B^{-1/8})$. $\square$

Corollary 7 (fixed windows). *Fix integers $1 \leq a \leq b$. Then*

$$\frac{1}{B} \sum_{q \leq B} \left(\tau(q; [1, a]) - \tau(q; (a, b]) \right) = 2H_a - H_b + O\left(\frac{b}{B}\right).$$

Proof. By Lemma 1, $\sum_{q \leq B} \tau(q; [1, m]) = \sum_{d \leq m} \lfloor B/d \rfloor = B H_m + O(m)$ for any fixed m ; take $m = a$ and $m = b$, subtract, and use $\tau(q; (a, b]) = \tau(q; [1, b]) - \tau(q; [1, a])$. $\square$

Remark 8. The constant γ enters Theorem 4 only through the absolute placement of a single threshold. Every statement about windows with both endpoints at fixed powers of q is γ -free, with the count governed, to first and second order, by the exponent length $b - a$ alone.

4. THE CONSTANT AS AN AVERAGE DISCREPANCY

Theorem 9. *Let $F : \mathbb{N} \rightarrow \mathbb{R}$ satisfy (a) F weakly increases; (b) $F(q) \rightarrow \infty$; (c) $q/F(q) \rightarrow \infty$. Put*

$$Z_F(q) = \#\{d \mid q : d^2 < F(q)\} - \#\{d \mid q : d^2 \geq F(q), d < F(q)\}.$$

Then

$$\frac{1}{B} \sum_{q \leq B} Z_F(q) \longrightarrow \gamma \quad (B \rightarrow \infty),$$

and the same limit obtains under any endpoint convention at $\sqrt{F(q)}$ and at $F(q)$.

Taking $F(q) = \sqrt{q}$: on average, a number has exactly γ more divisors below its fourth root than between its fourth root and its square root. Condition (c) cannot be dropped: for $F(q) = q$ the correspondence $d \leftrightarrow q/d$ pairs the divisors of q below $\sqrt{q}$ bijectively with those above, so $Z_F \equiv 0$.

The proof uses a counting inverse for F together with a lemma that disposes of boundary divisors.

Lemma 10 (boundary lemma). *Let $G : \mathbb{N} \rightarrow \mathbb{R}$ weakly increase, with $G(q) \rightarrow \infty$ and $G(B) = o(B)$. Then*

$$\#\{q \leq B : G(q) \in \mathbb{N} \text{ and } G(q) \mid q\} = o(B).$$

Proof. For each integer m the set $I_m = \{q : G(q) = m\}$ is an interval of integers (possibly empty), since G weakly increases, and the I_m are pairwise disjoint. Each q counted lies in I_m for $m = G(q)$ and is a multiple of m , so the count is at most $\sum_m (\#(I_m \cap [1, B])/m + 1)$, the sum running over the m with $I_m \cap [1, B] \neq \emptyset$. Fix m_0 . The terms with $m \leq m_0$ total at most $\#\{q : G(q) \leq m_0\} + m_0$, a bound finite and independent of B since $G \rightarrow \infty$. The terms with $m > m_0$ total at most

$$\frac{1}{m_0} \sum_m \#(I_m \cap [1, B]) + \#\{m : I_m \cap [1, B] \neq \emptyset\} \leq \frac{B}{m_0} + G(B) + 1.$$

Dividing by B , letting $B \rightarrow \infty$, and then letting $m_0 \rightarrow \infty$ gives the claim, since $G(B) = o(B)$. $\square$

Proof of Theorem 9. For $y \geq 0$ put $R(y) = \#\{n \in \mathbb{N} : F(n) \leq y\}$, finite by (b). Since F weakly increases,

$$(1) \quad F(n) \leq y \iff n \leq R(y).$$

Discarding the finitely many q with $F(q) < 1$ (an $O(1)$ adjustment to all sums), every divisor d with $d^2 < F(q)$ also satisfies $d < F(q)$, so

$$Z_F(q) = 2g(q) - t(q), \quad g(q) := \#\{d \mid q : d^2 < F(q)\}, \quad t(q) := \#\{d \mid q : d < F(q)\},$$

exactly. Write $T_1 = \sum_{q \leq B} g(q)$ and $T_2 = \sum_{q \leq B} t(q)$.

By Lemma 1 and (1),

$$T_1 = \#\{(d, e) : de \leq B, de > R(d^2)\}, \quad T_2 = \#\{(d, e) : de \leq B, de > R(d)\}.$$

In T_1 , the count of qualifying e for a given d vanishes unless $R(d^2) < B$, that is (by (1) applied to $n = B$), unless $d^2 < F(B)$; and when it does not vanish it equals $\lfloor B/d \rfloor - \lfloor R(d^2)/d \rfloor$. Hence, with $D_1 = \#\{d : d^2 < F(B)\}$ and $D_2 = \#\{d : d < F(B)\}$,

$$T_1 = \sum_{d \leq D_1} \left(\frac{B}{d} - \frac{R(d^2)}{d} \right) + O(D_1), \quad T_2 = \sum_{d \leq D_2} \left(\frac{B}{d} - \frac{R(d)}{d} \right) + O(D_2),$$

so that

$$(2) \quad 2T_1 - T_2 = B(2H_{D_1} - H_{D_2}) - S + O(F(B)), \quad S := 2 \sum_{d \leq D_1} \frac{R(d^2)}{d} - \sum_{m \leq D_2} \frac{R(m)}{m}.$$

Condition (c) makes the error term $o(B)$.

The harmonic bracket. By (b), $D_1 \rightarrow \infty$ and $D_2 \rightarrow \infty$, while $D_1 = \sqrt{F(B)} + O(1)$ and $D_2 = F(B) + O(1)$ give $D_1^2/D_2 \rightarrow 1$. Lemma 2 yields

$$2H_{D_1} - H_{D_2} = \gamma + \log \frac{D_1^2}{D_2} + O\left(\frac{1}{D_1}\right) \rightarrow \gamma,$$

so $B(2H_{D_1} - H_{D_2}) = \gamma B + o(B)$.

The sum S . We show $S = o(B)$. Every value $R(y)$ appearing in S has $y < F(B)$ and hence, by (1), $R(y) < B$. Group the second sum in S into blocks $m \in [d^2, (d+1)^2]$ for $1 \leq d \leq D_1 - 1$, together with a top segment $m \in [D_1^2, D_2]$. For the top segment,

$$\sum_{m=D_1^2}^{D_2} \frac{R(m)}{m} \leq B \left(\log \frac{D_2}{D_1^2} + O(D_1^{-2}) \right) = o(B),$$

again because $D_1^2/D_2 \rightarrow 1$. Within the block indexed by d , monotonicity of R traps each $R(m)$ between $R(d^2)$ and $R((d+1)^2 - 1)$, while

$$\sum_{m=d^2}^{(d+1)^2-1} \frac{1}{m} = \log \frac{(d+1)^2}{d^2} + O\left(\frac{1}{d^2}\right) = \frac{2}{d} + O\left(\frac{1}{d^2}\right),$$

so the block sum differs from $2R(d^2)/d$ by at most

$$\frac{2\Delta_d}{d} + O\left(\frac{R((d+1)^2 - 1)}{d^2}\right), \quad \Delta_d := R((d+1)^2 - 1) - R(d^2) \geq 0.$$

The first sum in S exceeds its truncation at $D_1 - 1$ by the single term $2R(D_1^2)/D_1 \leq 2B/D_1 = o(B)$. Hence

$$|S| \leq 2 \sum_{d \leq D_1-1} \frac{\Delta_d}{d} + O\left(\sum_{d \leq D_1-1} \frac{R((d+1)^2 - 1)}{d^2}\right) + o(B).$$

Fix d_0 and split each remaining sum at d_0 . The initial segments involve only the values $R(y)$ with $y \leq (d_0 + 1)^2$, constants independent of B , so they contribute $O_{d_0}(1)$. For the tails, the differences $\Delta_d \geq 0$ attach to the pairwise disjoint, increasingly ordered intervals $[d^2, (d+1)^2 - 1]$, so monotonicity of R gives $\sum_{d \leq D_1-1} \Delta_d \leq R(D_1^2 - 1) \leq B$; hence

$$\sum_{d_0 < d \leq D_1-1} \frac{\Delta_d}{d} \leq \frac{B}{d_0}, \quad \sum_{d_0 < d \leq D_1-1} \frac{R((d+1)^2 - 1)}{d^2} \leq B \sum_{d > d_0} \frac{1}{d^2} \leq \frac{B}{d_0}.$$

Therefore $\limsup_{B \rightarrow \infty} |S|/B \leq C/d_0$ for every d_0 , and $S = o(B)$.

Combining the estimates in (2) gives $(2T_1 - T_2)/B \rightarrow \gamma$, the assertion for the stated convention.

Endpoint conventions. At most one divisor of q can satisfy $d^2 = F(q)$, and it exists only when $\sqrt{F(q)} \in \mathbb{N}$ and $\sqrt{F(q)} \mid q$. Reassigning such divisors between the two windows therefore changes $\sum_{q \leq B} Z_F(q)$ by at most $2\#\{q \leq B : \sqrt{F(q)} \in \mathbb{N}, \sqrt{F(q)} \mid q\}$, which is $o(B)$ by Lemma 10 applied to $G = \sqrt{F}$: this G weakly increases, tends to infinity by (b), and satisfies $\sqrt{F(B)} = o(B)$ by (c). Likewise, admitting or barring the divisor $d = F(q)$ changes the total by at most $\#\{q \leq B : F(q) \in \mathbb{N}, F(q) \mid q\}$, which is $o(B)$ by Lemma 10 applied to $G = F$, using (c) directly. $\square$

Remark 11. For $F(q) = q^s$ with fixed $s \in (0, 1)$, Theorem 4 and Corollary 5 already contain Theorem 9, with a rate:

$$\frac{1}{B} \sum_{q \leq B} Z_F(q) = \left(\frac{s}{2} \log B + \gamma - \frac{s}{2} \right) - \frac{s}{2} (\log B - 1) + O(B^{-\delta}) = \gamma + O(B^{-\delta})$$

for some $\delta = \delta(s) > 0$. In the general theorem no rate can be uniform, since conditions (b) and (c) permit F to grow, or to shed relative growth, arbitrarily slowly.

Remark 12. Theorem 9 is convention-free precisely on account of (b) and (c): a divisor sitting at $\sqrt{F(q)}$ has size tending to infinity by (b), and a divisor sitting at $F(q)$ has codivisor $q/F(q)$ tending to infinity by (c), so both boundary populations thin to density zero. When (c) fails, the divisor at the top of the window retains positive density and the limit becomes convention-sensitive; the next section quantifies this.

5. WINDOWS ON THE SCALE OF q

Theorem 13. Fix $\alpha \in (0, 1)$ and set $K = \lfloor 1/\alpha \rfloor$ and $K' = \lceil 1/\alpha \rceil - 1 = \lceil (1 - \alpha)/\alpha \rceil$. Then

$$(3) \quad \frac{1}{B} \sum_{q \leq B} \left(\tau(q; [1, \sqrt{\alpha q}]) - \tau(q; (\sqrt{\alpha q}, \alpha q]) \right) \longrightarrow H_K - \log \frac{1}{\alpha},$$

$$(4) \quad \frac{1}{B} \sum_{q \leq B} \left(\tau(q; [1, \sqrt{\alpha q}]) - \tau(q; (\sqrt{\alpha q}, \alpha q]) \right) \longrightarrow H_{K'} - \log \frac{1}{\alpha}.$$

The indices K and K' agree unless $1/\alpha$ takes an integer value k , in which case $K = k$ and $K' = k - 1$, and the two limits differ by $1/k = \alpha$.

Proof. Write

$$g(q) = \tau(q; [1, \sqrt{\alpha q}]), \quad t(q) = \tau(q; [1, \alpha q]), \quad \beta(q) = \#\{d \mid q : d = \sqrt{\alpha q}\},$$

$$\rho(q) = \#\{d \mid q : d = \alpha q\}.$$

Splitting $[1, \alpha q]$ at the point $\sqrt{\alpha q}$ shows that the summand in (3) equals $2g(q) - t(q) + \beta(q)$, and the summand in (4) equals $2g(q) - t(q) + \beta(q) - \rho(q)$.

The boundary counts. For $q = de$, the equation $d = \sqrt{\alpha q}$ reads $d^2 = \alpha de$, that is, $d = \alpha e$. If α is irrational this has no solutions and $\beta \equiv 0$. If $\alpha = s/t$ in lowest terms, then $td = se$ with $\gcd(s, t) = 1$ forces $e = tm$, $d = sm$, $q = stm^2$ for some m , so $\sum_{q \leq B} \beta(q) \leq \sqrt{B} = o(B)$. Next, $d = \alpha q$ reads $e = 1/\alpha$, so $\rho \equiv 0$ unless $1/\alpha$ is an integer k , in which case $\rho(q) = 1$ exactly when $k \mid q$, and $\sum_{q \leq B} \rho(q) = \lfloor B/k \rfloor = \alpha B + O(1)$.

The count below $\sqrt{\alpha q}$. For $q = de$, $d < \sqrt{\alpha q} \iff d^2 < \alpha de \iff e > d/\alpha$. By Lemma 1 and the same reasoning as in Theorem 4 (the count over e vanishes unless $d < \sqrt{\alpha B}$),

$$\sum_{q \leq B} g(q) = \sum_{d < \sqrt{\alpha B}} \left(\frac{B}{d} - \frac{d}{\alpha} \right) + O(\sqrt{B}) = B H_{\lfloor \sqrt{\alpha B} \rfloor - 1} - \frac{1}{\alpha} \cdot \frac{\alpha B}{2} + O(\sqrt{B}),$$

so that Lemma 2 gives

$$\frac{1}{B} \sum_{q \leq B} g(q) = \frac{\log B}{2} + \frac{\log \alpha}{2} + \gamma - \frac{1}{2} + O(B^{-1/2}).$$

The count below αq . For $q = de$, $d \geq \alpha q \iff \alpha e \leq 1 \iff e \leq K$ (for integer e). Hence $t(q) = \tau(q) - \#\{e \mid q : e \leq K\}$, and Lemmas 1 and 3 give

$$\frac{1}{B} \sum_{q \leq B} t(q) = \log B + 2\gamma - 1 - H_K + O(B^{-1/2}).$$

Assembly.

$$\frac{1}{B} \sum_{q \leq B} (2g(q) - t(q)) = 2 \left(\frac{\log B}{2} + \frac{\log \alpha}{2} + \gamma - \frac{1}{2} \right) - (\log B + 2\gamma - 1 - H_K) + o(1) = H_K + \log \alpha + o(1).$$

Adding the $o(1)$ contribution of β gives (3). Subtracting the contribution of ρ gives (4): for $1/\alpha \notin \mathbb{Z}$ nothing changes and $K' = K$; for $1/\alpha = k$ the limit drops by $\alpha = 1/k$, and $H_k - 1/k - \log k = H_{k-1} - \log k = H_{K'} - \log(1/\alpha)$. $\square$

Remark 14. Write $A(\alpha) = H_{\lfloor 1/\alpha \rfloor} - \log(1/\alpha)$ for the limit in (3); the limit in (4) differs only at the points $\alpha = 1/k$. On each interval $(\frac{1}{k+1}, \frac{1}{k}]$ the function A agrees with the continuous, strictly increasing function $\alpha \mapsto H_k + \log \alpha$; as α decreases through $1/k$ the value jumps up by $1/k$, the average newly crediting, for one q in every k , the divisor q/k to the window. Version (3) takes the upper value at the jump and version (4) the lower, so the two versions are, respectively, the left-continuous and right-continuous regularizations of a single function. By Lemma 2, $A(\alpha) - \gamma = \theta_k - \log(1/(k\alpha))$ for $\alpha \in (\frac{1}{k+1}, \frac{1}{k}]$ with $0 < \theta_k < 1/k$, whence $A(\alpha) \rightarrow \gamma$ as $\alpha \rightarrow 0^+$, consistently with Theorem 9, whose hypothesis (c) the choice $F(q) = \alpha q$ just barely fails. Since $\theta_k < \log \frac{k+1}{k}$, each branch crosses the level γ exactly once, at $\alpha = e^{\gamma - H_k} \in (\frac{1}{k+1}, \frac{1}{k})$; the crossings accumulate at 0, and γ is the only value that A attains infinitely often.

Proposition 15 (mean value). $\int_0^1 \left(H_{\lfloor 1/\alpha \rfloor} - \log \frac{1}{\alpha} \right) d\alpha = \zeta(2) - 1 = \frac{\pi^2}{6} - 1.$

Proof. On $(\frac{1}{k+1}, \frac{1}{k}]$ the integrand's first term is the constant H_k , so

$$\int_0^1 H_{\lfloor 1/\alpha \rfloor} d\alpha = \sum_{k \geq 1} \frac{H_k}{k(k+1)}.$$

The partial sums telescope:

$$\sum_{k=1}^N H_k \left(\frac{1}{k} - \frac{1}{k+1} \right) = H_1 + \sum_{j=2}^N \frac{H_j - H_{j-1}}{j} - \frac{H_N}{N+1} = 1 + \sum_{j=2}^N \frac{1}{j^2} - \frac{H_N}{N+1} \longrightarrow \zeta(2).$$

Since $\int_0^1 \log(1/\alpha) d\alpha = 1$, the proposition follows. (The integrands attached to (3) and (4) differ only on a null set.) $\square$

Remark 16 (numerical illustration). At $B = 4 \cdot 10^5$: for $F(q) = \sqrt{q}$ the average of Z_F over $q \leq B$ equals 0.57770, against $\gamma = 0.57722$; for $\alpha = 1/2$ the averages in (3) and (4) equal 0.80685 and 0.30685, against $H_2 - \log 2 = 0.80685$ and $1 - \log 2 = 0.30685$.

6. A LOGARITHM-FREE APPROXIMATION AND ITS RATE

Corollary 7 attaches to each pair $a \leq b$ the discrepancy $2H_a - H_b$, a quantity free of logarithms; choosing b near a^2 sends it to γ . The sharpest elementary choice places the

window $(n, n^2]$ against $[1, n)$, crediting the divisor $d = n$ to neither side:

$$\gamma_n := \sum_{k=1}^{n-1} \frac{1}{k} - \sum_{k=n+1}^{n^2} \frac{1}{k} = H_{n-1} + H_n - H_{n^2} = \lim_{B \rightarrow \infty} \frac{1}{B} \sum_{q \leq B} \left(\tau(q; [1, n)) - \tau(q; (n, n^2]) \right).$$

Proposition 17. $\gamma - \gamma_n = \frac{2}{3n^2} + O(n^{-4})$.

Proof. Since $H_{n-1} = H_n - 1/n$, the expansion of Lemma 2 gives

$$\gamma_n = 2H_n - \frac{1}{n} - H_{n^2} = 2\left(\log n + \gamma + \frac{1}{2n} - \frac{1}{12n^2}\right) - \frac{1}{n} - \left(2\log n + \gamma + \frac{1}{2n^2}\right) + O(n^{-4}),$$

and the right side collapses to $\gamma - (\frac{1}{6} + \frac{1}{2})n^{-2} + O(n^{-4}) = \gamma - \frac{2}{3}n^{-2} + O(n^{-4})$. $\square$

Remark 18. The symmetric choice, $2H_{n-1} - H_{n^2}$ or $2H_n - H_{n^2}$, approaches γ only at the rate $1/n$; excluding the single divisor $d = n$ from both windows cancels the first-order error. The sequence (γ_n) increases to γ , furnishing a definition of γ by harmonic sums alone: $\gamma = \lim_n (\sum_{k < n} 1/k - \sum_{n < k \leq n^2} 1/k)$, with error below n^{-2} .

7. FORMALIZATION

The principal results of this paper have been checked by machine, in Lean 4 against the Mathlib library [3], with the source archived at [4]. Verified: the counting identity underlying Lemma 1, in the form $\sum_{q \leq B} \#\{d \mid q : d \leq a\} = \sum_{d \leq a} \lfloor B/d \rfloor$; the fixed-window law of Corollary 7; the limit $2H_{\lfloor \sqrt{T} \rfloor} - H_T \rightarrow \gamma$ supplying the harmonic bracket in the proof of Theorem 9; Theorem 9 itself in the instance $F(q) = \sqrt{q}$, by way of Remark 11; Theorem 13 at $\alpha = 1/2$ in *both* endpoint conventions, so that the discrepancy of $1/2$ between the two constants is itself a verified statement; the vanishing of the window difference in Corollary 6; and the rate of Proposition 17. Each carries a clean axiom audit, depending only on the three axioms of Lean’s standard foundation.

The formal development follows the organization of the proofs above. A counting layer supplies the divisor–codivisor correspondence and the splitting of divisor counts at successive powers. Above it sits a single master asymptotic: writing $S(c, j, B) = \sum_d (\lfloor B/d \rfloor - c d^{j-1})$, truncated where the summand vanishes,

$$\frac{S(c, j, B)}{B} - \left(\frac{1}{j} (\log B - \log c) + \gamma - \frac{1}{j} \right) \longrightarrow 0 \quad (c \geq 1, j \geq 2),$$

which is Theorem 4 in the shape the applications need. Each verified limit is then a linear combination of instances of this asymptotic, and the cancellations described in Remark 8 appear formally as the vanishing of the corresponding weight sums: for Corollary 6 at exponents $j = 2, 4, 8$ with weights $1, -3, 2$, both $\frac{1}{2} - \frac{3}{4} + \frac{2}{8} = 0$ and $1 - 3 + 2 = 0$, killing the logarithm and the constant together.

The one result whose formal proof departs substantially from the argument given above is Proposition 17, where the Euler–Maclaurin expansion of Lemma 2 was replaced by an explicit two-sided bound: with $y_m = H_m - \log m - \frac{1}{2m} + \frac{1}{12m^2}$, the sequences $y_m \mp 10m^{-3}$ are respectively monotone and antitone from $m = 2$ onward and both converge to γ , trapping γ between them and yielding $|y_m - \gamma| \leq 10m^{-3}$; this gives the effective form $|n^2(\gamma - \gamma_n) - \frac{2}{3}| \leq 31/n$ for $n \geq 2$, of which Proposition 17 is the limiting case.

8. QUESTIONS

Question 19. Fix $\alpha \in (0, 1)$. By Theorem 13 the discrepancy between the divisor counts in $[1, \sqrt{\alpha n})$ and in $(\sqrt{\alpha n}, \alpha n)$ has positive average. Do the integers n for which the discrepancy is strictly positive have positive lower density? A positive average does not settle this: integers with many more divisors below $\sqrt{\alpha n}$ than above could in principle carry the whole average, although integers with $\tau(n)$ large compared with $\log n$ are rare.

Question 20. For $F(q) = q^s$, Remark 11 bounds the deviation of the average of Z_F from γ by a negative power of B . Determine the exact second-order asymptotics; the divisor-pair counts involved tie the question to the error term in the Dirichlet divisor problem.

Question 21. Beyond its mean, does $Z_F(q)$, suitably normalized, possess a limiting distribution as q ranges over $[1, B]$? The second moment already involves correlations among divisor counts in separated ranges.

REFERENCES

- [1] C.-J. de la Vallée Poussin, Sur les valeurs moyennes de certaines fonctions arithmétiques, *Ann. Soc. Sci. Bruxelles* **22** (1898), 84–90.
- [2] S. R. Finch, *Mathematical Constants*, Encyclopedia of Mathematics and its Applications 94, Cambridge University Press, Cambridge, 2003.
- [3] The mathlib Community, The Lean mathematical library, *CPP 2020: Proceedings of the 9th ACM SIG-PLAN International Conference on Certified Programs and Proofs*, 367–381.
- [4] D. V. Feldman, **gamma-divisors**: formalization companion to *Divisors in windows and the Euler–Mascheroni constant*, Zenodo, 2026, <https://doi.org/10.5281/zenodo.21909843>.
- [5] R. L. Graham, D. E. Knuth, and O. Patashnik, *Concrete Mathematics*, second edition, Addison-Wesley, Reading, MA, 1994.

DEPARTMENT OF MATHEMATICS AND STATISTICS, UNIVERSITY OF NEW HAMPSHIRE, DURHAM, NH 03824